Local and Latent: Dual Views of Symmetry Breaking in Cooperative Multi-Robot RL

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Abstract Spatial symmetry is among the most effective inductive biases for robot learning, but in real multi-robot deployments it almost never holds exactly. Violations come in two structurally distinct forms: *local* breakers, visible in the current state but confined to small regions (e.g., fixed obstacles), and *latent* breakers, hidden from observation but fixed across an episode (e.g., wind or gravity). The two are orthogonal—state-dependent gating cannot find an unobserved breaker, and hidden-context inference cannot represent localized agent-dependent violations—yet realistic deployments exhibit both. We propose a dual-view theory and unified architecture. View 1 formalizes a locally factored partially symmetric Markov game and shows that critic-level symmetry error is controlled by discounted future breaker exposure, motivating a TD-learned per-agent gated critic. View 2 formalizes a latent symmetry-breaker POMDP and bounds value error by breaker-estimation error. The unified architecture pairs an equivariant latent encoder with per-agent gating, yielding testable theoretical bounds, a practical multi-robot RL algorithm, and benchmarks that isolate local, latent, and mixed symmetry-breaking regimes.
Keywords cooperative multi-robot RL, group equivariance, symmetry breaking

1 Introduction

Group symmetries—geometric transformations leaving system properties invariant—provide one of the most effective inductive biases in robot learning, with consistent gains in imitation learning [1, 2, 3] and reinforcement learning (RL) [4, 5, 6], where exploration is expensive and demonstrations are scarce. Formally, given a group G acting on the states and actions of an MDP, the MDP is G-symmetric if its reward and transition are invariant under the group action. The notion extends to multi-robot settings by lifting actions and invariance to joint spaces, including PI's prior work [7, 8], where the relevant group factors as $G = G_{\text{spatial}} \times \mathrm{S}_n$. We focus on the spatial component and leave permutation symmetry (S_n) for future work.

In practice, symmetry almost never holds exactly. We observe that violations fall into two structurally distinct modes, separated by a single diagnostic: *is the breaker present in an agent's current observation, or hidden from it? Local* breakers are observable and confined to small regions of the per-agent state-action space, e.g., for warehouse robots near a fixed shelving unit, symmetry breaks only in the unit's vicinity, and at any moment only a subset of agents is affected. *Latent* breakers are hidden global variables fixed within an episode, e.g., wind on aerial drones, gravity on inclined multi-arm setups, current on underwater robots, where dynamics would be exactly symmetric *if* the vector were rotated together with state and action, yet the vector is not directly observable.

The two modes are structurally orthogonal: local methods cannot detect hidden breakers; a global latent variable is a poor inductive bias for representing agent-dependent localized violations. Realistic deployments exhibit both. This *local-latent* dichotomy separates our proposal from prior work, which either treats single-agent settings where the dichotomy does not arise or globally relaxes symmetry without distinguishing structurally different modes [9, 10, 11, 12, 13, 14, 15]. Closing this gap is the goal of this proposal.

2 Methods

We consider a discounted cooperative Markov game with agent set $\mathcal{N} = \{1, \ldots, N\}$, global state space $\mathcal{S} = \mathcal{S}_1 \times \cdots \times \mathcal{S}_N$ decomposed into per-agent local states, joint action space $\mathcal{A} = \mathcal{A}_1 \times \cdots \times \mathcal{A}_N$,

and team reward $r\colon \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, equipped with a spatial group G lifted to each agent's local state and action via group actions. We pursue two complementary views of why G may fail to be exact: View 1 attributes failures to localized regions of the observed local state-action space, motivating a state-dependent gated critic; View 2 attributes them to an unobserved global breaker variable (wind, gravity, current) and recovers equivariance in the augmented state through breaker inference. The two will later be combined in a single architecture.

2.1 View 1: symmetry breakers as locally observed state-action regions

We posit a *locally factored* partially symmetric Markov game in which the reward and transition decompose agent-wise: $r(s,a) = \sum_{i=1}^{N} r_i(s_i,a_i), P(\cdot|s,a) = \prod_{i=1}^{N} P_i(\cdot|s_i,a_i)$. Each local factor carries its own finite symmetry group G acting on $(\mathcal{S}_i, \mathcal{A}_i)$ and its own *local breaker region* $\mathcal{B}_i \subseteq \mathcal{S}_i \times \mathcal{A}_i$. On the safe set $(s_i,a_i) \notin \mathcal{B}_i$ the local reward and transition are G-invariant; on $\mathcal{B}_i$ they may be arbitrary. The breaker region is itself G-invariant, so the safe/broken designation is a property of the orbit. This is more general than a single global gate over the joint state: symmetry can be exploited at agents whose local context is symmetric even when others' are not. We introduce a locally routed model with routing parameter $\lambda_i(s_i,a_i) \in [0,1]$. Let $\overline{r}_i$ and $\overline{P}_i$ denote the G-orbit-averaged reward and transition. The routed factor interpolates between the equivariant and raw models: $\widetilde{r}_i^{\lambda} = (1-\lambda_i)\overline{r}_i + \lambda_i r_i$ and $\widetilde{P}_i^{\lambda} = (1-\lambda_i)\overline{P}_i + \lambda_i P_i$. Thus $\lambda_i = 0$ recovers the fully equivariant model, while $\lambda_i = 1$ recovers the raw model. The oracle breaker route $\lambda_i^*(s_i,a_i) := \mathbf{1}\{(s_i,a_i) \in \mathcal{B}_i\}$ is useful as a diagnostic, but our analysis and algorithm allow arbitrary learned routes.

Theory. For any bounded local critic Q_i, the routed model has a one-step Bellman error bounded by the residual of the equivariant approximation, scaled by the amount of averaging used, i.e., $|(\widetilde{\mathcal{T}}_i^{\lambda} Q_i)(s_i,a_i) - (\mathcal{T}_i Q_i)(s_i,a_i)| \lesssim (1-\lambda_i(s_i,a_i))(\Delta_i^r + \gamma V_{i,\max}\Delta_i^P)$. Thus small λ_i exploits equivariance when local residuals are small, while large λ_i avoids bias near local breakers. We will use this one-step bound to show that local routing controls the induced joint-critic and policy-gradient errors without requiring a single global gate. At the critic level, the key point is that a locally safe pair $(s_i,a_i) \notin \mathcal{B}_i$ can still have asymmetric value if future trajectories enter a breaker region. We will formalize this through a discounted future-exposure term $\beta_i^{\pi_i}(s_i,a_i)$ and show that critic asymmetry is controlled by this exposure, motivating gates learned from TD error rather than hand-coded breaker indicators. Thus, the learned gate should not simply detect whether the current local state-action pair lies in a breaker region; it should estimate whether the current pair is likely to lead to asymmetric future values.

Algorithm. Algorithmically, this view leads to a per-agent routed critic. Each agent i uses a gate $\lambda_i(s_i,a_i) \in [0,1]$ to interpolate between an equivariant local critic and an unconstrained local critic: $Q_i^{\lambda} = (1-\lambda_i)Q_i^{\mathrm{eq}} + \lambda_i Q_i^{\mathrm{free}}$. We will learn λ_i from TD error and counterfactual consistency signals, allowing each agent to exploit symmetry in locally safe regimes while routing around observable symmetry breakers.

2.2 View 2: symmetry breakers as partially observed latent state

For hidden-context breakers, the agent-wise local-region view does not fit: the breaker is a single global object (e.g., a wind or gravity vector) shared by all agents and fixed across the episode. Throughout this subsection we therefore use single-agent-style notation (s,a,o,b), with the understanding that joint quantities decompose per-agent and G acts simultaneously on all agents' frames; agent-wise factorization returns in §2.3. We model the environment as a *latent symmetry-breaker POMDP* in which breaker $b \in \mathcal{B}$ is sampled at the start of each episode and held fixed; the agent receives observations $o_t \sim \Omega_b(\cdot|s_t)$ and history h_t but never sees b. The structural assumption is *augmented equivariance*: for every $g \in G$, $r_{g\cdot b}(g\cdot s, g\cdot a) = r_b(s,a)$, $\quad P_{g\cdot b}(g\cdot C|g\cdot s, g\cdot a) = P_b(C|s,a)$, $\quad \Omega_{g\cdot b}(g\cdot D|g\cdot s) = \Omega_b(D|s)$. The system is symmetric if and only if b is transformed

alongside (s, a); the visible state alone admits no group action consistent with the dynamics.

Theory. We will establish two key results that drive the algorithm. First, under Lipschitz assumptions $|r_b - r_{b'}| \leq L_r d_{\mathcal{B}}(b, b')$ and $\|P_b - P_{b'}\|_{\mathrm{TV}} \leq L_p d_{\mathcal{B}}(b, b')$, conditioning on an estimate $\widehat{b}$ rather than the true b yields a fixed-point value error bounded by $\frac{L_r + \gamma V_{\max} L_p}{1-\gamma} d_{\mathcal{B}}(b, \widehat{b})$, motivating auxiliary objectives that drive $d_{\mathcal{B}}(b, \widehat{b})$ down. Second, under augmented equivariance and an equivariant prior ρ_0 over $\mathcal{B}$, the Bayesian posterior over b transforms equivariantly: $\beta_t^{g \cdot h_t}(g \cdot B) = \beta_t^{h_t}(B)$; the encoder f_η should therefore be built so its output transforms in the same representation as b.

Algorithm. The architecture has three components. A latent breaker encoder $z_t = f_\eta(h_t)$ (recurrent or contrastive) is built so $f_\eta(g \cdot h_t) = g \cdot f_\eta(h_t)$, i.e., its output lives in the same representation as b. A breaker-conditioned policy and critic are equivariant in the augmented pair (o, z): $\pi_\theta(g \cdot a | g \cdot o, g \cdot z) = \pi_\theta(a | o, z)$ and $Q_\omega(g \cdot o, g \cdot a, g \cdot z) = Q_\omega(o, a, z)$. The encoder is trained with auxiliary objectives in addition to TD: a dynamics loss $\mathcal{L}_{\mathrm{dyn}} = \mathbb{E}[\ell(D_\psi(o_t, a_t, z_t), o_{t+1})]$, a contrastive loss pulling same-episode representations together (and pushing across-episode pairs apart), and an optional supervised loss $\mathcal{L}_{\mathrm{sup}} = \mathbb{E}[d_{\mathcal{B}}(f_\eta(h_t), b)^2]$ when simulator labels are available. We will explore three variants: a point-estimate encoder producing $\widehat{b}_t \in \mathcal{B}$; a belief representation outputting distributional parameters; and canonicalization, rotating inputs into a frame aligned with $\widehat{b}_t$, applying a standard network, and rotating outputs back.

2.3 Unifying perspective and combined architecture

The two views compose because they operate at different granularities: View 1 is per-agent and state-dependent, while View 2 is global and per-episode. We therefore replace View 1's orbit-averaged branch with View 2's breaker-conditioned equivariant branch, using a shared latent encoder $z_t = f_\eta(h_t)$ that is broadcast to all agents, while retaining the per-agent gate. Each agent's critic is $Q_i(s_i, a_i, h_t) = \big(1 - \lambda_{i,\psi}(s_i, a_i)\big) q_{i,\theta}^{\mathrm{cond}}(s_i, a_i, z_t) + \lambda_{i,\psi}(s_i, a_i) q_{i,\phi}^{\mathrm{raw}}(s_i, a_i, h_t)$, and the joint critic is $Q(s, a, h_t) = \sum_i Q_i(s_i, a_i, h_t)$. The breaker-conditioned branch exploits symmetry after accounting for hidden global context, while the raw branch handles localized observable violations. Showing that this combined architecture outperforms both single-view variants on mixed-breaker tasks is a primary empirical claim.

Evaluation. We will evaluate three regimes designed to isolate the two failure modes and their interaction. First, multi-robot navigation with fixed obstacles isolates agent-wise local breakers. Second, hidden-wind navigation removes obstacles but introduces a per-episode wind vector omitted from observations, isolating latent global breakers. Third, multi-arm manipulation on an inclined surface with localized obstacles combines hidden gravity with local observed violations. Metrics include final return, sample efficiency, training stability across seeds, and view-specific diagnostics such as learned gates, latent-breaker estimation error, counterfactual equivariance error, etc.

3 Expected results: key outcomes and timeline

The project spans 14 months. The first three months focus on setup and baselines implementation. View 1 theory and algorithms are developed in months 2–6, running in parallel with View 2 theory and algorithms in months 4–8. Unification of the two views begins in month 8 and continues through month 11. The final three months are dedicated to code release, paper writing, and submission.

4 Funds needed

Cash funding: \$42,450 is requested for the annual stipend of a research assistant (RA) [16]. *AWS promotional credits:* \$50,000 is requested. ∼\$35,000 will be allocated to EC2 for parallelized multi-robot RL training, ∼\$10,000 to RoboMaker and SageMaker to enable realistic and large-scale robotics simulation, training, and benchmarking, ∼\$5,000 to supporting infrastructure.

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